

FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING

Department of Computer Engineering

Academic Year 2025-2026

Rubrics for Lab Experiments

Class: B.E. Computer
Engineering

Subject Name: SMA Lab

Semster: VIII

Subject Code: CSDL8023

Practical No.	6
Title:	Develop Structure based social media analytics model for any business. (e.g. Structure Based Models-community detection, influence analysis)
Date of Performance:	1-12-2025
Roll No.	9913
Student	Mark Lopes

Evaluation

Performance Indicator	Below Average	Average	Good	Excellent	Marks
On Time Submission (2)	Not submitted(0)	Submitted after deadline (1)	Early or on time submission (2)	—	
Test cases and output (4)	Incorrect output (1)	The expected output is verified only a for few test cases (2)	The expected output is Verified for all test cases but is not presentable (3)	Expected output is obtained for all test cases. Presentable and easy to follow (4)	
Coding efficiency (2)	The code is not structured at all (0)	The code is structured but not efficient (1)	The code is Structured and efficient. (2)	-	
Knowledge(2)	Basic concepts not clear (0)	Understood the basic concepts (1)	Could explain the concept with suitable example (1.5)	Could relate the theory with real world application(2)	
Total					

Signature of the Teacher:

Aim: Develop Structure based social media analytics model for any business. (e.g. Structure Based Models-community detection, influence analysis)

Code:

./main.py

```
pip install networkx pandas matplotlib

import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt
import os

csv_path = '/content/pseudo_facebook.csv'
df = pd.read_csv(csv_path)

df.head(7)

# --- ADAPTING DATA FOR EXPERIMENT 6 ---

# Since 'pseudo_facebook.csv' is demographic data, not a graph,
# we will simulate a graph by assuming users with similar ages are 'connected'
# or strictly based on the first column 'userid'.

# Let's create a synthetic edge list from the first 100 users for visualization
# (Connecting users if they are within the same age group or sequentially)

print("--- Transforming Data to Network Format ---")

# Taking a subset to avoid crashing the graph visualizer
subset = df.head(100)

# Create an empty graph
G = nx.Graph()

# Add nodes (users)
for index, row in subset.iterrows():
    G.add_node(row['userid'], age=row['age'], gender=row['gender'])

# Create Edges (Simulation: Connect user i to user i+1 to form a chain/network)
user_ids = subset['userid'].tolist()
for i in range(len(user_ids) - 1):
    G.add_edge(user_ids[i], user_ids[i+1])
    # Randomly add another connection to make it interesting
    if i % 5 == 0:
        G.add_edge(user_ids[i], user_ids[(i+5) % len(user_ids)])

print(f"Graph Created with {G.number_of_nodes()} nodes and {G.number_of_edges()} edges.")

# 1. Centrality (Influence Analysis)
print("\n--- Centrality Measures ---")
degree_cent = nx.degree_centrality(G)
top_5 = sorted(degree_cent.items(), key=lambda x: x[1], reverse=True)[:5]
print("Top 5 Influential Users (by connection):", top_5)
```

```

# 2. Clustering (Community Detection)
print("\n--- Community Detection ---")
communities = list(nx.community.greedy_modularity_communities(G))
print(f"Communities detected: {len(communities)}")

# Map for coloring
community_map = {}
for i, comm in enumerate(communities):
    for node in comm:
        community_map[node] = i

# 3. Visualization
plt.figure(figsize=(10, 8))
colors = [community_map.get(node, 0) for node in G.nodes()]
pos = nx.spring_layout(G, seed=42)

nx.draw_networkx(G, pos,
                 node_color=colors,
                 node_size=300,
                 cmap=plt.cm.viridis,
                 with_labels=False)
plt.title("Facebook User Network (Simulated Connections)")
plt.show()

```

Output:

Facebook User Network (Simulated Connections)

